

OmniTrade Hub

AGILE SPRING PLANNING

GROUP 10

Overview and Methodology

This sprint planning document outlines the adjusted execution roadmap for OmniTrade Hub, leading up to our final presentation on August 25, 2026. Following requirement analysis and database architecture design, the project is structured into four milestone sprints to ensure core features, secure checkout transactions, and testing are fully completed.

Milestone Sprint Roadmap

Sprint 1: Architecture Setup & Database Foundation

Goal: Establish the decoupled three-tier solution structure and deploy the SQL Server database schema via Entity Framework Core.

Initialize `OmniTradeApi`, `OmniTradeMvc`, and `OmniTradeTests` (xUnit) projects. Configure `OmniTradeDbContext` for all relational entities (`Users`, `Vendors`, `Products`, `CartItems`, `Orders`, `OrderItems`).

Sprint 2: Authentication & Catalog Management

Goal: Deliver user registration workflows and complete CRUD operations for merchant product catalogs.

Build authentication and role-based endpoints in the Web API. Develop the `ProductsController` and MVC views allowing vendors to add, update, and delete product listings.

Sprint 3: Cart & Checkout Transaction Engine

Goal: Implement the buyer shopping cart and the atomic checkout transaction pipeline with multi-vendor order grouping.

Build cart management views enabling buyers to add items from multiple vendors into a unified cart. Implement checkout transaction endpoint: generate order references, group items by vendor, deduct stock quantities atomically, and set status to Pending.

Sprint 4: Fulfilment Pipeline & Presentation Prep (Deadline: Aug 25)

Goal: *Finalize order state-transitions, admin dashboards, and dry-run presentation review.*

Implement order fulfilment lifecycle states (Pending -> Processing -> Shipped -> Delivered). Conduct integration testing and finalize presentation slides for the August 25 deadline.

Task Board & Member Distribution

Member	Assigned Focus Area	Key Deliverables
Aaditya	Project Lifecycle & Architecture	Solution scaffolding, project structure setup, and lifecycle documentation.
Abhishek	Authentication & Users API	`Users` table schema, seeding, and secure login/register endpoints.
Harsh	Vendor Management Module	`Vendors` table schema, merchant profile registration, and approval flags.
Anshul	Product Catalog Module	`Products` table schema, catalog CRUD APIs, and filtering views.
Shreyas	Shopping Cart Integration	`CartItems` schema, cart endpoints, and unified buyer cart UI.
Lakshya	Checkout & Transaction Engine	`Orders` schema, atomic stock deduction, and order generation logic.
Kushagra	Order Items & Fulfillment Pipeline	`OrderItems` schema, multi-vendor order splitting, and status state machine.